Psychiatry Review

Schizophrenia • Major depressive disorder • Bipolar disorder • Anxiety • Insomnia • Substance use disorders — all key points for the exam, with a drug table after each topic

1. Schizophrenia

Diagnosis (DSM-5-TR)

- **≥ 2 symptoms for ≥ 1 month** (at least one of the first three): **delusions**, **hallucinations** (auditory most common), **disorganized speech**, grossly disorganized or **catatonic** behavior, **negative symptoms**.
- Continuous signs of disturbance for **≥ 6 months** • rule out substances, mood disorders, medical causes.
- **Positive symptoms** (delusions, hallucinations) respond best to antipsychotics • **negative** (flat affect, alogia, avolition, anhedonia, social withdrawal) and **cognitive** symptoms respond poorly.
- **Catatonia** → IV/IM **lorazepam** (or ECT).

Dopamine pathways and antipsychotic effects

Pathway	Role	Effect of D2 blockade
Mesolimbic	↑ dopamine → **positive symptoms**	**Therapeutic** (↓ hallucinations, delusions)
Mesocortical	↓ dopamine → negative/cognitive symptoms	May **worsen** negative symptoms
Nigrostriatal	Movement	**EPS**, tardive dyskinesia
Tuberoinfundibular	Dopamine inhibits prolactin	**Hyperprolactinemia** (galactorrhea, amenorrhea, gynecomastia, sexual dysfunction)

Antipsychotic classes

- **First generation (typical, FGA):** D2 antagonists • **high potency** (haloperidol, fluphenazine) → **more EPS** and prolactin • **low potency** (chlorpromazine, thioridazine) → more **sedation**, **anticholinergic**, **orthostatic hypotension**, less EPS.
- **Second generation (atypical, SGA):** D2 + **5-HT2A antagonism** → less EPS and TD • more **metabolic** effects (some).
- **D2 partial agonists: aripiprazole**, **brexpiprazole**, **cariprazine** → low prolactin, low metabolic risk, but **akathisia**.
- **Xanomeline-trospium** (Cobenfy): **muscarinic M1/M4 agonist** (no D2 blockade) → no EPS, no weight gain, no prolactin rise.
- **Efficacy is similar** for all antipsychotics (except **clozapine**, which is superior in treatment-resistant schizophrenia) → choose by **side-effect profile**, comorbidities, prior response, formulation.

Side-effect comparison (high-yield)

Drug	EPS	Prolactin	Metabolic	Sedation	QT	Ortho.	Anti-chol.
Haloperidol	+++	+++	+	+	+ (IV ++)	+	+
Chlorpromazine	+	++	++	+++	++	+++	+++
Clozapine	0	0	**+++**	+++	+	**+++**	**+++**
Olanzapine	+	+	**+++**	++	+	+	++
Quetiapine	0/+	0	++	**+++**	+	++	+
Risperidone	++ (dose)	**+++**	++	+	+	++	0
Paliperidone	++	**+++**	++	+	+	++	0
Ziprasidone	+	+	**0/+**	+	**++**	+	0
Lurasidone	++	+	**0/+**	+	0	+	0
Aripiprazole	+ (**akathisia**)	**0 (↓)**	**0/+**	+	0	+	0
Iloperidone	0/+	+	++	+	++	**+++**	+

- **Ortho.** = orthostatic hypotension • **Anti-chol.** = anticholinergic effects.
- **Highest metabolic risk: clozapine** and **olanzapine** • **lowest: aripiprazole**, **ziprasidone**, **lurasidone**, brexpiprazole, cariprazine, lumateperone.
- **Highest prolactin: risperidone**, **paliperidone**, high-potency FGAs.

Specific SGA points

Drug	Key points
Risperidone	EPS at doses > 6 mg/day • highest prolactin among SGAs • LAIs available
Paliperidone	Active metabolite of risperidone • **renal elimination** (adjust) • LAIs monthly, 3-monthly, 6-monthly
Olanzapine	Weight gain, diabetes, dyslipidemia • **olanzapine/samidorphan** (Lybalvi) ↓ weight gain (**CI with opioids**) • olanzapine pamoate LAI → **post-injection delirium/sedation syndrome** (observe 3 h; REMS) • CYP1A2 (smoking ↓ levels)
Quetiapine	Very **sedating**, orthostasis, low EPS (good for PD psychosis) • cataracts (eye exams) • bipolar depression • misuse potential
Ziprasidone	**Take with food (≥ 500 kcal)** • **QT prolongation** (avoid with other QT drugs)
Lurasidone	**Take with food (≥ 350 kcal)** • **CI with strong CYP3A4 inhibitors and inducers** • bipolar depression
Asenapine	**Sublingual** — no food or drink for 10 min • oral numbness (hypoesthesia), dysgeusia • patch form
Iloperidone	Slow titration (orthostatic hypotension) • QT
Aripiprazole / brexpiprazole / cariprazine	**Akathisia** • low prolactin and metabolic risk • aripiprazole: **impulse-control disorders** (gambling) • cariprazine: very long half-life • used as MDD augmentation (aripiprazole, brexpiprazole, cariprazine)
Lumateperone	Low metabolic/EPS risk • bipolar depression • sedation
Xanomeline-trospium (Cobenfy)	Take **1 h before or 2 h after** meals • ADRs: **nausea, vomiting, constipation, dyspepsia**, **urinary retention**, tachycardia • **CI:** urinary retention, hepatic impairment, gastric retention, narrow-angle glaucoma

Clozapine (treatment-resistant schizophrenia)

- **Indication: treatment-resistant** schizophrenia (failure of **≥ 2 adequate antipsychotic trials**) • **↓ suicidal behavior** • most effective antipsychotic.
- **Severe neutropenia** (agranulocytosis) → **ANC monitoring** (baseline, then weekly × 6 months, every 2 weeks × 6 months, then monthly) — the national REMS program was **removed in 2025**, but ANC monitoring is still recommended.

- **Boxed warnings: severe neutropenia**, **orthostatic hypotension/bradycardia/syncope** (slow titration), **seizures** (dose-related, > 600 mg/day), **myocarditis/cardiomyopathy** (first 1–2 months: chest pain, fever, tachycardia → troponin, CRP), **↑ mortality in elderly dementia**.
- Other: **severe constipation/ileus** (bowel regimen), **sialorrhea** (glycopyrrolate, atropine drops SL), **weight gain, diabetes**, sedation, fever, enuresis.
- **Smoking induces CYP1A2** → ↓ clozapine levels • **quitting smoking → ↑ levels** (reduce dose ~30–50%, check level) • **fluvoxamine, ciprofloxacin ↑ levels** • target level ≥ 350 ng/mL.
- If **≥ 2 days (48 h) of doses missed** → **re-titrate** from 12.5 mg.

Treatment principles

- **Adequate trial:** therapeutic dose for **2–6 weeks** (some improvement in 1–2 weeks; full effect may take months).
- **First episode:** continue **≥ 1–2 years** after remission • **multiple episodes:** long-term/indefinite.
- **Non-adherence** → **long-acting injectable (LAI)** (establish tolerability with oral form first).
- **Monitoring:** weight/BMI (baseline, monthly × 3, then quarterly), waist, BP, **glucose/A1c and lipids** (baseline, 12 weeks, then yearly), **EPS and AIMS** (TD), prolactin if symptomatic, ECG (QT drugs), eye exam (quetiapine), CBC (clozapine).
- **Pregnancy:** generally continue the effective antipsychotic.

Long-acting injectables (LAIs)

LAI	Frequency / notes
Haloperidol decanoate	Every 4 weeks (deep IM, Z-track)
Fluphenazine decanoate	Every 2–3 weeks
Risperidone (Risperdal Consta / Perseris / Uzedy / Rykindo)	Consta every 2 weeks (**needs 3 weeks oral overlap**) • Perseris monthly SC • Uzedy SC monthly or every 2 months (no oral overlap)
Paliperidone palmitate	Sustenna: **234 mg day 1, 156 mg day 8 (deltoid)**, then monthly (no oral overlap) • Trinza every 3 months (after ≥ 4 months of Sustenna) • Hafyera every 6 months
Aripiprazole (Maintena monthly, Asimtufii every 2 months, Aristada)	Maintena/Asimtufii: 14 days oral overlap (or 2-injection start) • Aristada: 21 days oral or Aristada Initio + 30 mg oral once
Olanzapine pamoate (Zyprexa Relprevv)	Every 2–4 weeks • **post-injection delirium sedation syndrome** → observe 3 hours (REMS)

Managing antipsychotic adverse effects

Adverse effect	Onset / features	Management
Acute dystonia	Hours–days • neck twisting, oculogyric crisis, laryngospasm • young men, high-potency FGAs	**Diphenhydramine 25–50 mg** or **benztropine 1–2 mg IM/IV** → continue oral for a few days
Akathisia	Days–weeks • inner restlessness, can't sit still	**Reduce dose**, **propranolol 10–30 mg BID–TID**, benzodiazepine, low-dose mirtazapine
Pseudoparkinsonism	Weeks • tremor, rigidity, bradykinesia	**Benztropine**, trihexyphenidyl, **amantadine**; reduce dose or switch
Tardive dyskinesia (TD)	Months–years • involuntary orofacial movements; may be permanent	**Valbenazine** or **deutetrabenazine** (VMAT2 inhibitors) • switch to clozapine/quetiapine • **anticholinergics worsen TD**
Neuroleptic malignant syndrome (NMS)	Days • **fever**, **"lead-pipe" rigidity**, **altered mental status**, **autonomic instability**, ↑ **CK**, myoglobinuria	**Stop antipsychotic**, supportive care (cooling, fluids), **dantrolene**, **bromocriptine**, benzodiazepines
Metabolic	Weight gain, diabetes, dyslipidemia	Lifestyle, switch to lower-risk drug, **metformin**
Hyperprolactinemia	Galactorrhea, amenorrhea, gynecomastia, sexual dysfunction	Lower dose, switch, or **add aripiprazole**
QT prolongation	Torsades risk (ziprasidone, iloperidone, IV haloperidol, thioridazine)	ECG, correct K/Mg, avoid other QT drugs
Orthostatic hypotension	Clozapine, iloperidone, quetiapine, low-potency FGAs	Slow titration, rise slowly
Anticholinergic	Dry mouth, constipation, urinary retention, confusion	Avoid in elderly; bowel regimen

Boxed warning (all antipsychotics): ↑ mortality in elderly patients with dementia-related psychosis (stroke, infection) — not approved for this use (except brexpiprazole for Alzheimer's agitation).

Schizophrenia drug summary

Drug	Usual adult dose	Key ADRs	Key DDIs / notes
Haloperidol	2–20 mg/day (acute agitation 2–10 mg IM)	**EPS**, prolactin, QT (IV)	CYP2D6/3A4
Chlorpromazine	200–800 mg/day	Sedation, orthostasis, anticholinergic, ↓ seizure threshold	Low potency
Fluphenazine	2.5–20 mg/day	EPS	Decanoate LAI
Risperidone	2–6 mg/day	**Prolactin**, EPS (> 6 mg)	CYP2D6 (paroxetine, fluoxetine ↑)
Paliperidone	3–12 mg/day	Prolactin, EPS, QT	**Renal adjustment**
Olanzapine	10–20 mg/day	**Weight gain, diabetes**, sedation	**CYP1A2**: smoking ↓, fluvoxamine ↑
Quetiapine	300–800 mg/day	**Sedation**, orthostasis, weight gain, cataracts	CYP3A4 inhibitors ↑
Ziprasidone	40–80 mg BID **with food**	**QT**	Avoid other QT drugs
Lurasidone	40–160 mg daily **with food**	Akathisia, sedation	**CI strong CYP3A4 inhibitors/inducers**
Asenapine	5–10 mg SL BID	Oral numbness, sedation	No food/drink for 10 min
Iloperidone	Titrate to 6–12 mg BID	**Orthostasis**, QT	CYP2D6/3A4
Aripiprazole	10–30 mg/day	**Akathisia**, impulse-control disorders	CYP2D6/3A4 (halve dose with strong inhibitors)
Brexpiprazole	2–4 mg/day	Akathisia, weight gain	CYP2D6/3A4
Cariprazine	1.5–6 mg/day	**Akathisia**, EPS	CYP3A4 • long half-life
Lumateperone	42 mg daily	Sedation, dry mouth	CYP3A4 inducers avoid
Xanomeline-trospium	50/20 mg BID → 125/30 mg BID (empty stomach)	**Nausea, vomiting, constipation, urinary retention**, tachycardia	CI urinary retention, glaucoma, hepatic impairment
Clozapine	12.5 mg → 300–450 mg/day (max 900)	**Neutropenia**, **seizures**, **myocarditis**, orthostasis, constipation, sialorrhea, weight gain	**CYP1A2**: smoking ↓, **quitting smoking ↑**, fluvoxamine/cipro ↑ • ANC monitoring
Benztropine	1–2 mg PO/IM/IV BID	Anticholinergic	Dystonia, parkinsonism (not TD)
Diphenhydramine	25–50 mg IM/IV	Sedation, anticholinergic	Acute dystonia
Propranolol	10–30 mg BID–TID	Bradycardia, hypotension	Akathisia • avoid in asthma
Valbenazine	40 mg → 80 mg daily	Somnolence, QT	TD • strong CYP3A4 inhibitors → 40 mg • CI MAOIs
Deutetrabenazine	6 mg BID → max 48 mg/day with food	Somnolence, **depression/suicidality** (boxed in Huntington's)	TD • CYP2D6 • CI MAOIs, reserpine
Dantrolene	1–2.5 mg/kg IV (NMS)	Hepatotoxicity, weakness	NMS
Bromocriptine	2.5 mg TID (NMS)	Nausea, hypotension	NMS

2. Major Depressive Disorder (MDD)

Diagnosis

- **≥ 5 symptoms for ≥ 2 weeks**, including **depressed mood** or **anhedonia** (loss of interest) — mnemonic **SIG E CAPS**: **S**leep, **I**nterest, **G**uilt, **E**nergy, **C**oncentration, **A**ppetite, **P**sychomotor changes, **S**uicidal ideation.
- **PHQ-9**: 5–9 mild, 10–14 moderate, 15–19 moderately severe, ≥ 20 severe.

- **Screen for bipolar disorder** before starting an antidepressant (antidepressant monotherapy can trigger **mania**).
- **Rule out:** hypothyroidism, substance use, medical illness, and **drug-induced depression** (interferons, **isotretinoin**, corticosteroids, reserpine, efavirenz, levetiracetam, perampanel, montelukast, hormonal contraceptives, varenicline (rare)).
- **Always assess suicide risk.**

Treatment principles

- **Psychotherapy (CBT)** and/or antidepressants for mild–moderate • **combination** for moderate–severe.
- **First-line drugs: SSRIs**, **SNRIs**, **bupropion**, **mirtazapine** (similar efficacy) → choose by **side effects**, **DDIs**, comorbidities, prior response, family response, cost.
- **Onset:** some improvement (sleep, energy) in 1–2 weeks • full effect **4–8 weeks** • adequate trial = **4–8 weeks at a therapeutic dose**.
- **Goal: remission** (not just response ≥ 50% improvement).
- **Duration: continuation 4–9 months after remission** (first episode) • **maintenance ≥ 2 years or lifelong** if ≥ 3 episodes, chronic, or severe.
- **Stopping: taper over ≥ 2–4 weeks** → avoid **discontinuation syndrome** (**FINISH**: Flu-like, Insomnia, Nausea, Imbalance, Sensory disturbances like "brain zaps", Hyperarousal) — worst with **paroxetine** and **venlafaxine**, least with **fluoxetine**.
- **Not responding:** optimize dose → **switch** (same or different class) or **augment**.
- **Severe, psychotic, catatonic, or high suicide risk: ECT** • psychotic depression → antidepressant + antipsychotic.

> **Boxed warning (all antidepressants): ↑ suicidal thoughts and behavior** in children, adolescents, and young adults **< 25 years** → close monitoring, especially in the **first weeks** and after dose changes.

Antidepressant classes

Class / drugs	Key ADRs	Key notes / DDIs
SSRIs: citalopram, escitalopram, fluoxetine, paroxetine, sertraline, fluvoxamine	**GI** (nausea, diarrhea), **sexual dysfunction**, insomnia or sedation, headache, **hyponatremia/SIADH** (elderly), **bleeding** (with NSAIDs/anticoagulants), bone loss, weight gain (long term), apathy, bruxism	**Citalopram: QT** → max **40 mg** (**20 mg** if > 60 years, CYP2C19 poor metabolizer, hepatic impairment, or with cimetidine) • **fluoxetine:** long half-life (norfluoxetine), activating, **strong CYP2D6 inhibitor** • **paroxetine:** most **anticholinergic**, sedating, weight gain, **worst discontinuation**, **strong CYP2D6 inhibitor** (↓ **tamoxifen** activation), **avoid in pregnancy** (cardiac defects) • **sertraline:** diarrhea; preferred in cardiac disease and pregnancy • **fluvoxamine: strong CYP1A2 inhibitor** (**CI with tizanidine**, ↑ theophylline, clozapine)
SNRIs: venlafaxine, desvenlafaxine, duloxetine, levomilnacipran	SSRI effects + **↑ blood pressure** (venlafaxine, dose-related), sweating, tachycardia, urinary hesitancy, dry mouth	**Duloxetine:** neuropathic pain, fibromyalgia, GAD • **hepatotoxicity** (avoid heavy alcohol, liver disease) • avoid CrCl < 30 • CYP1A2/2D6 • **venlafaxine:** worst discontinuation • **desvenlafaxine:** renal adjustment, few DDIs
Bupropion (NDRI)	Insomnia, dry mouth, tremor, ↑ BP, **seizures** (dose-related)	**No sexual dysfunction**, **no weight gain**, activating • also **smoking cessation** • **CI: seizure disorder**, **bulimia/anorexia**, abrupt alcohol/benzodiazepine/antiseizure drug withdrawal, **MAOIs** • max 450 mg/day • **strong CYP2D6 inhibitor**
Mirtazapine (α2 antagonist)	**Sedation** (more at low doses), **↑ appetite and weight gain**, dry mouth, rare agranulocytosis	**No sexual dysfunction** • good for **insomnia, poor appetite, elderly** • ODT form
Trazodone	**Sedation**, **orthostatic hypotension**, **priapism**, QT	Mostly used at low dose (25–100 mg) for **insomnia**
Nefazodone	**Boxed: hepatotoxicity**	Rarely used • CYP3A4 inhibitor
Vilazodone (SSRI + 5-HT1A partial agonist)	**Diarrhea**, nausea	**Take with food** • titrate 10 → 20 → 40 mg • CYP3A4 (max 20 mg with strong inhibitors)
Vortioxetine (multimodal)	**Nausea**, constipation	Less sexual dysfunction • may improve cognition • CYP2D6
TCAs: amitriptyline, nortriptyline, imipramine, desipramine, clomipramine, doxepin	**Anticholinergic**, **sedation**, **orthostasis**, weight gain, **cardiac conduction** (QRS widening, arrhythmias)	**Lethal in overdose** (treat with **sodium bicarbonate**) → limit quantities in suicidal patients • **avoid in elderly** (Beers) • ECG if > 40 or cardiac disease • nortriptyline level 50–150 ng/mL • also neuropathic pain, migraine prevention • clomipramine: OCD
MAOIs: phenelzine, tranylcypromine, isocarboxazid, **selegiline patch**	**Orthostatic hypotension**, weight gain, sexual dysfunction, insomnia	**Hypertensive crisis** with **tyramine** (aged cheese, cured/smoked meats, soy sauce, tap beer, sauerkraut) and **sympathomimetics** (pseudoephedrine, phenylephrine, amphetamines) • **serotonin syndrome** with SSRIs, SNRIs, TCAs, meperidine, tramadol, methadone, **dextromethorphan**, **linezolid**, **methylene blue**, St. John's wort • **washout 2 weeks** when switching (**5 weeks after fluoxetine**) • selegiline patch 6 mg/24 h: no diet restriction

Augmentation and newer options

Agent	Key points
Second-generation antipsychotics	**Aripiprazole** 2–15 mg, **brexpiprazole** 0.5–3 mg, **quetiapine XR** 150–300 mg, **cariprazine** 1.5 mg, **olanzapine/fluoxetine** (treatment-resistant) • watch akathisia, metabolic effects
Lithium	Augmentation (levels ~0.6–1.0 mEq/L) • ↓ suicide
Liothyronine (T3)	25–50 mcg/day
Combination antidepressants	e.g., SSRI/SNRI + bupropion or mirtazapine
Esketamine nasal (Spravato)	**Treatment-resistant depression** (≥ 2 failed antidepressants) and MDD with suicidal ideation, with an oral antidepressant • 56–84 mg twice weekly × 4 weeks, then weekly, then every 1–2 weeks • **REMS**: given in a certified clinic, **observe 2 hours** (sedation, **dissociation**, ↑ BP) • no driving that day • Schedule III (abuse)
Dextromethorphan-bupropion (Auvelity)	45/105 mg daily × 3 days → BID • faster onset (~1 week) • seizures, dizziness • CYP2D6 • serotonin syndrome
Zuranolone (Zurzuvae)	**Postpartum depression • 50 mg once daily in the evening with fatty food × 14 days • boxed: CNS depression → no driving for 12 hours** after each dose • CYP3A4
Brexanolone	Postpartum depression • **60-hour IV infusion** • excessive sedation/loss of consciousness (REMS, pulse oximetry)
ECT / TMS	ECT: most effective for severe, psychotic, suicidal, catatonic depression • TMS: treatment-resistant

Serotonin syndrome vs neuroleptic malignant syndrome

Feature	Serotonin syndrome	NMS
Cause	Serotonergic drugs (combinations, overdose, MAOIs)	Dopamine antagonists (or stopping dopamine agonists)
Onset	**Rapid (within 24 h)**	**Slow (days)**
Neuromuscular	**Clonus**, **hyperreflexia**, tremor, myoclonus	**"Lead-pipe" rigidity**, bradyreflexia
Other	Agitation, diaphoresis, hyperthermia, diarrhea, mydriasis	Fever, altered mental status, autonomic instability, **↑ CK**
Treatment	Stop drugs, supportive care, **benzodiazepines**, **cyproheptadine** 12 mg then 2 mg q2h	Stop drug, supportive, **dantrolene**, **bromocriptine**

Special populations

Population	Key points
Pregnancy	**Sertraline**, citalopram/escitalopram, fluoxetine acceptable • **avoid paroxetine** (cardiac defects) • 3rd-trimester SSRIs → neonatal adaptation syndrome, PPHN (small risk) • untreated depression is also risky
Breastfeeding	**Sertraline** and paroxetine (low milk levels) preferred
Elderly	**Sertraline**, **escitalopram**, citalopram ≤ 20 mg, **mirtazapine** (insomnia, weight loss) • **avoid TCAs, paroxetine** • watch **hyponatremia**, falls, bleeding
Children/adolescents	**Fluoxetine (≥ 8 years)** and **escitalopram (≥ 12 years)** FDA-approved for MDD • boxed suicidality warning
Comorbidities	Pain/neuropathy → duloxetine, TCAs • smoking → bupropion • sexual dysfunction → bupropion, mirtazapine, vortioxetine • insomnia/weight loss → mirtazapine • seizures → avoid bupropion • cardiac → sertraline • hypertension → avoid venlafaxine high doses
Other	**St. John's wort**: CYP3A4/P-gp **inducer** (↓ OCs, cyclosporine, warfarin, HIV drugs) + serotonin syndrome risk • seasonal depression → light therapy, bupropion XL • persistent depressive disorder (≥ 2 years)

MDD drug summary

Drug	Usual adult dose	Key ADRs	Key DDIs / notes
Citalopram	20–40 mg (max 20 mg if > 60)	**QT**, GI, sexual	CYP2C19 • cimetidine → max 20 mg
Escitalopram	10–20 mg	GI, sexual, QT (less)	Few DDIs • ≥ 12 years
Fluoxetine	20–80 mg (weekly 90 mg)	Activation, insomnia	**Strong CYP2D6 inhibitor** • 5-week MAOI washout • ≥ 8 years
Paroxetine	20–50 mg	**Anticholinergic**, weight gain, sedation, **discontinuation**	**Strong CYP2D6 inhibitor** (tamoxifen) • avoid pregnancy, elderly
Sertraline	50–200 mg	**Diarrhea**, GI	Preferred: cardiac, pregnancy, breastfeeding
Fluvoxamine	50–300 mg (OCD)	GI, sedation	**Strong CYP1A2 inhibitor** (tizanidine CI, theophylline, clozapine)
Venlafaxine XR	75–225 mg	↑ **BP**, sweating, **discontinuation**	CYP2D6
Desvenlafaxine	50 mg daily	Same	Renal adjustment
Duloxetine	30–60 mg (max 120)	Nausea, **hepatotoxicity**	Avoid heavy alcohol, CrCl < 30 • CYP1A2 (fluvoxamine CI), CYP2D6
Levomilnacipran	40–120 mg	↑ HR, BP, urinary hesitancy	CYP3A4 inhibitors → max 80 mg
Bupropion XL	150 → 300–450 mg AM	**Seizures**, insomnia	**CI seizures, eating disorders, MAOIs** • CYP2D6 inhibitor
Mirtazapine	15–45 mg HS	**Sedation, weight gain**	Elderly with insomnia/poor appetite
Trazodone	Insomnia 25–100 mg HS • MDD 150–400 mg	Sedation, **priapism**, orthostasis	CYP3A4
Vilazodone	10 → 20 → 40 mg **with food**	Diarrhea, nausea	Strong CYP3A4 inhibitors → max 20 mg
Vortioxetine	5–20 mg	Nausea	CYP2D6 poor metabolizers → max 10 mg
Amitriptyline	25 → 150–300 mg HS	Anticholinergic, sedation, **cardiotoxic**	Beers • lethal overdose
Nortriptyline	25 → 75–150 mg	Less anticholinergic	Level 50–150 ng/mL
Phenelzine / tranylcypromine	15 mg TID (phenelzine) • 10 mg BID–TID (tranylcypromine)	Orthostasis, weight gain	**Tyramine**, sympathomimetics, serotonergic drugs • 2-week washout
Selegiline patch	6–12 mg/24 h	Application-site reactions	No tyramine diet at 6 mg
Esketamine nasal	56–84 mg twice weekly → weekly → q1–2 wk	Dissociation, sedation, ↑ BP	REMS, observe 2 h
Dextromethorphan-bupropion	45/105 mg daily × 3 d → BID	Dizziness, seizures	CYP2D6 • serotonergic
Zuranolone	50 mg evening with fat × 14 days	**CNS depression** (boxed)	No driving 12 h • CYP3A4
Aripiprazole (augment)	2–15 mg/day	Akathisia	CYP2D6/3A4
Cyproheptadine	12 mg then 2 mg q2h (serotonin syndrome)	Sedation, anticholinergic	Antidote

3. Bipolar Disorder

Diagnosis and types

- **Manic episode:** abnormally elevated, expansive, or irritable mood + ↑ goal-directed activity for **≥ 1 week** (any duration if hospitalized), with **≥ 3 symptoms** (**4 if only irritable**) — **DIGFAST**: Distractibility, Impulsivity/high-risk activities, Grandiosity, Flight of ideas, Activity/agitation, Sleep deficit (↓ need), Talkativeness (pressured speech).
- **Hypomania:** same features for **≥ 4 days**, **no** hospitalization, **no** marked impairment, **no** psychosis.
- **Bipolar I:** ≥ 1 **manic** episode • **Bipolar II:** hypomania + major depression (never full mania) • **Cyclothymia:** ≥ 2 years of subthreshold symptoms • **rapid cycling:** ≥ 4 mood episodes per year.
- **Drug-induced mania: antidepressants** (monotherapy), **corticosteroids**, stimulants, levodopa, testosterone/anabolic steroids.

Treatment by phase

Phase	First-line options	Notes
Acute mania / mixed	**Lithium**, **valproate**, or an **SGA** (aripiprazole, asenapine, cariprazine, olanzapine, quetiapine, risperidone, ziprasidone) • haloperidol • severe: **lithium or valproate + SGA**	**Stop antidepressants** • benzodiazepines for agitation/insomnia • ECT • valproate preferred for mixed and rapid cycling
Bipolar depression	**Quetiapine**, **lurasidone**, **cariprazine**, **lumateperone**, **olanzapine/fluoxetine** • lithium • lamotrigine (maintenance)	**Avoid antidepressant monotherapy** (switch to mania, rapid cycling) — if used, always with a mood stabilizer
Maintenance	**Lithium** (gold standard; ↓ suicide), **valproate**, **lamotrigine** (prevents **depressive** episodes; not useful for acute mania), SGAs (aripiprazole, quetiapine, olanzapine, risperidone LAI, asenapine)	Continue the regimen that achieved stabilization • often long-term/lifelong

Lithium (high-yield)

Point	Details
Dose	Start **300 mg BID–TID** → titrate to level • once-daily dosing at bedtime possible for maintenance
Target levels	**Acute mania 0.8–1.2 mEq/L** • **maintenance 0.6–1.0 mEq/L** • draw a **trough 12 hours** after the last dose, **5 days** after a dose change (steady state)
Toxicity	**> 1.5:** GI (nausea, vomiting, diarrhea), **coarse tremor**, drowsiness • **> 2.0:** confusion, ataxia, slurred speech, muscle twitching • **> 2.5:** seizures, coma, arrhythmias, renal failure → **hemodialysis** (severe or with kidney failure)
Adverse effects	**Fine tremor** (propranolol) • **polyuria/polydipsia** → **nephrogenic diabetes insipidus** (amiloride) • **hypothyroidism** • weight gain • GI upset • **acne**, psoriasis worsening • cognitive dulling • benign leukocytosis • ECG T-wave changes • **CKD** (long term) • **hyperparathyroidism/hypercalcemia** • **teratogen**: **Ebstein's anomaly**
Monitoring	**Lithium levels**, **SCr/BUN**, **electrolytes**, **TSH**, **calcium**, weight, ECG (> 40 or cardiac), pregnancy test • every 3–6 months once stable
↑ Lithium levels	**Thiazide diuretics** (most important), **ACE inhibitors**, **ARBs**, **NSAIDs** (not aspirin/sulindac), **dehydration**, **low-sodium diet**, vomiting/diarrhea, metronidazole, renal impairment
↓ Lithium levels	**Caffeine**, **theophylline**, **high sodium intake**, osmotic diuretics, acetazolamide, **pregnancy** (↑ clearance)
Counseling	**Consistent salt and fluid intake** • stay hydrated (heat, exercise, illness) • recognize toxicity • avoid OTC NSAIDs • take with food to ↓ GI upset

Other mood stabilizers

- **Valproate (divalproex):** mania loading **20–30 mg/kg/day** • level **50–125 mcg/mL** • good for **mixed episodes and rapid cycling** • boxed: hepatotoxicity, pancreatitis, **teratogen** (avoid in women of childbearing potential) • weight gain, tremor, alopecia, thrombocytopenia, hyperammonemia • ↑ lamotrigine levels.
- **Lamotrigine:** maintenance, **prevents depression** • target **200 mg/day** • slow titration (SJS) • **halve with valproate**, ↑ with inducers, **estrogen OCs ↓ levels**.
- **Carbamazepine ER** (Equetro): acute mania • SJS (HLA-B*1502), aplastic anemia, hyponatremia • strong inducer.

Pregnancy and special populations

- **Avoid valproate** (worst) and **carbamazepine** in pregnancy • **lithium**: small Ebstein's risk → fetal echo; **levels fall in pregnancy** (↑ dose) and rise after delivery (reduce dose near delivery) • **lamotrigine** and many **SGAs** are reasonable options.
- Monitor **metabolic parameters** for SGAs (weight, glucose, lipids).
- **Elderly:** lower lithium levels (↓ renal clearance), more toxicity.

Bipolar drug summary

Drug	Usual adult dose	Key ADRs	Key DDIs / notes
Lithium	300 mg BID–TID → level 0.6–1.2 mEq/L	Tremor, polyuria (nephrogenic DI), **hypothyroidism**, weight gain, acne, CKD, **Ebstein's**	**↑ by thiazides, ACEi/ARBs, NSAIDs**, dehydration • ↓ by caffeine, theophylline, high sodium
Divalproex	Load 20–30 mg/kg/day → level 50–125	**Hepatotoxicity, pancreatitis, teratogen**, weight gain, tremor, ↓ platelets, ↑ NH3	↑ lamotrigine • carbapenems ↓ VPA
Lamotrigine	25 mg × 2 wk → 50 mg × 2 wk → 100 mg × 1 wk → 200 mg/day	**SJS**	Halve with VPA • estrogen ↓
Carbamazepine ER	200 mg BID → max 1,600 mg/day	SJS, aplastic anemia, hyponatremia	Strong inducer
Quetiapine	Mania 400–800 mg • depression 300 mg HS	Sedation, metabolic	CYP3A4
Lurasidone	20–120 mg daily with food (bipolar depression)	Akathisia	CI strong 3A4 inhibitors/inducers
Cariprazine	Mania 3–6 mg • depression 1.5–3 mg	Akathisia	CYP3A4
Lumateperone	42 mg daily (bipolar depression)	Sedation	CYP3A4
Olanzapine/fluoxetine	6/25 → 12/50 mg HS	Weight gain, sedation	Bipolar depression, TRD
Aripiprazole	15–30 mg/day	Akathisia	Mania, maintenance
Asenapine	5–10 mg SL BID	Oral numbness	No food/drink 10 min
Amiloride	5–10 mg BID	Hyperkalemia	Lithium-induced nephrogenic DI

4. Anxiety Disorders

Types

Disorder	Key features
Generalized anxiety disorder (GAD)	Excessive worry on most days for **≥ 6 months** + **≥ 3 of 6**: restlessness, fatigue, poor concentration, irritability, **muscle tension**, sleep disturbance
Panic disorder	Recurrent **unexpected panic attacks** (peak within minutes: palpitations, sweating, chest pain, fear of dying) + ≥ 1 month of worry or avoidance • ± agoraphobia
Social anxiety disorder	Fear of social or performance situations (scrutiny) • **performance-only** subtype
PTSD	After trauma, **> 1 month**: intrusions (flashbacks, **nightmares**), avoidance, negative mood/cognition, hyperarousal
OCD	Obsessions (intrusive thoughts) and/or compulsions (repetitive behaviors)
Specific phobia	Fear of a specific object/situation → **exposure therapy (CBT)**

- **Drugs/substances causing anxiety: caffeine**, **stimulants**, decongestants (pseudoephedrine), **beta-agonists**, theophylline, **corticosteroids**, thyroid hormone excess, bupropion, SSRI initiation • **withdrawal** from alcohol, benzodiazepines, nicotine.
- Rule out medical causes: hyperthyroidism, arrhythmias, hypoglycemia, pheochromocytoma, asthma/COPD.

First-line treatment

- **CBT** and/or **SSRIs/SNRIs** = first-line for **GAD, panic, social anxiety, PTSD**.
- **Start low** (half the depression starting dose, especially in **panic disorder** → avoid initial jitteriness) and titrate.
- Onset **2–4 weeks**, full effect **8–12 weeks** • continue **~12 months** after response, then taper slowly.

Disorder	Preferred drugs	Other options
GAD	**SSRIs** (escitalopram, paroxetine, sertraline), **SNRIs** (duloxetine, venlafaxine XR)	**Buspirone**, pregabalin, hydroxyzine, benzodiazepines (short-term)
Panic disorder	**SSRIs** (sertraline, paroxetine, fluoxetine, escitalopram), **venlafaxine XR**	Benzodiazepines (short-term bridge: clonazepam, alprazolam) • TCAs • **buspirone NOT effective**
Social anxiety	**SSRIs** (paroxetine, sertraline), **venlafaxine XR**	**Performance anxiety only**: **propranolol 10–40 mg** 30–60 min before the event
PTSD	**Sertraline**, **paroxetine** (FDA-approved), venlafaxine, fluoxetine + trauma-focused psychotherapy	**Prazosin** for **nightmares** • **avoid benzodiazepines** in PTSD
OCD	**SSRIs at higher doses** (fluoxetine 60–80, sertraline 200, fluvoxamine 300 mg) • longer trial (**8–12 weeks**)	**Clomipramine** • augment with **aripiprazole** or risperidone

Buspirone (GAD only)

- **5-HT1A partial agonist** • **7.5 mg BID** → 15–30 mg BID (max 60 mg/day).
- **Delayed onset (2–4 weeks)** → **not for PRN use** • **not effective for panic**, social anxiety, PTSD.
- **No sedation, no dependence, no withdrawal, no abuse** • ADRs: dizziness, nausea, headache.
- **DDIs: CYP3A4** (grapefruit, azoles, macrolides ↑) • **MAOIs** (hypertension).

Benzodiazepines

- **Role: short-term** (2–4 weeks) **bridge** while an SSRI starts working, or PRN for acute severe anxiety • **not long-term monotherapy**.
- **Boxed warnings: risks with opioids** (profound sedation, respiratory depression, death) • **abuse, misuse, addiction** • **dependence and withdrawal**.
- **ADRs:** sedation, **cognitive impairment**, ataxia, **falls** (elderly — Beers), anterograde amnesia, paradoxical agitation, respiratory depression.

- **Withdrawal:** rebound anxiety, insomnia, tremor, **seizures** → **taper slowly** (e.g., ↓ 10–25% every 1–2 weeks) • worse with short-acting agents.
- **Avoid in:** substance use history, **elderly**, respiratory disease (sleep apnea, COPD), pregnancy, PTSD.
- **Overdose antidote: flumazenil** (may cause **seizures** in chronic users).

Benzodiazepine	Half-life / features
Alprazolam	Short • rapid onset • **interdose rebound**, high abuse potential • panic • CYP3A4 (azoles CI)
Lorazepam	Intermediate • **no active metabolites** (glucuronidation) → OK in **liver disease and elderly** (LOT: Lorazepam, Oxazepam, Temazepam) • IV/IM forms
Oxazepam / temazepam	Intermediate • LOT • temazepam for insomnia
Clonazepam	Long • panic disorder, seizures • less rebound
Diazepam	Long with **active metabolites** • alcohol withdrawal, muscle spasm, status epilepticus • accumulates in elderly
Chlordiazepoxide	Long • **alcohol withdrawal**

Anxiety drug summary

Drug	Usual adult dose	Key ADRs	Key DDIs / notes
Escitalopram	5–10 mg → 20 mg	GI, sexual	GAD
Sertraline	25 mg → 50–200 mg	GI, diarrhea	Panic, social, PTSD, OCD
Paroxetine	10 mg → 20–60 mg	Anticholinergic, sedation	CYP2D6 inhibitor • avoid pregnancy
Fluvoxamine	50 → 300 mg (OCD)	GI, sedation	**CYP1A2 inhibitor** (tizanidine CI)
Venlafaxine XR	37.5 → 75–225 mg	↑ BP, sweating	GAD, panic, social
Duloxetine	30 → 60–120 mg	Nausea, hepatotoxicity	GAD
Buspirone	7.5 mg BID → 15–30 mg BID	Dizziness, nausea	GAD only • CYP3A4 • MAOIs
Hydroxyzine	25–50 mg up to QID	Sedation, anticholinergic, QT	Beers • PRN
Pregabalin	150–600 mg/day	Dizziness, edema	Off-label GAD (US) • renal
Propranolol	10–40 mg 30–60 min before event	Bradycardia, fatigue	Performance anxiety • avoid asthma
Prazosin	1 mg HS → 3–15 mg	**First-dose orthostatic hypotension**	PTSD nightmares
Clomipramine	25 → 100–250 mg	Anticholinergic, seizures, sexual	OCD
Alprazolam	0.25–0.5 mg TID (panic up to 4–10 mg/day)	Sedation, dependence, rebound	**CYP3A4 inhibitors** (ketoconazole CI)
Lorazepam	0.5–2 mg BID–TID	Sedation, dependence	LOT
Clonazepam	0.25–0.5 mg BID → 1–4 mg/day	Sedation, dependence	Panic
Diazepam	2–10 mg BID–QID	Sedation, accumulation	Active metabolites
Flumazenil	0.2 mg IV, repeat (max 1 mg)	**Seizures** (chronic users)	Benzodiazepine reversal

5. Insomnia

Overview

- Difficulty **initiating** or **maintaining** sleep (or early awakening) with daytime impairment • **chronic**: ≥ 3 nights/week for ≥ 3 months • **short-term**: < 3 months.
- **Look for causes:** pain, GERD, nocturia, COPD/asthma, heart failure, depression, anxiety, sleep apnea, restless legs, **drugs**.
- **Drugs causing insomnia: caffeine**, stimulants, **decongestants**, **SSRIs/SNRIs**, **bupropion**, **corticosteroids**, **beta-agonists**, theophylline, levothyroxine excess, **diuretics at night**, beta-blockers (nightmares), **alcohol** (fragmented sleep), nicotine, selegiline.

Non-drug therapy (first-line)

- **CBT for insomnia (CBT-I)** = **first-line** and most effective long term.
- **Sleep hygiene:** regular schedule, avoid caffeine/alcohol/nicotine late, avoid screens, comfortable dark quiet room, exercise (not late), avoid heavy meals at night.
- **Stimulus control** (bed only for sleep; leave bed if not asleep in ~20 min) • **sleep restriction** • relaxation techniques.

Drug therapy (AASM)

- Use the **lowest effective dose** for the **shortest time**, together with CBT-I • choose by sleep problem (**onset** vs **maintenance**).
- **Sleep onset: zaleplon**, **ramelteon**, zolpidem, eszopiclone, (triazolam, temazepam).
- **Sleep maintenance: suvorexant**/other DORAs, **low-dose doxepin**, **zolpidem ER**, eszopiclone, temazepam.
- **Middle-of-the-night awakening: low-dose sublingual zolpidem** (only if ≥ 4 hours left in bed).
- **Not recommended** (AASM): **trazodone**, **diphenhydramine**/doxylamine, **melatonin**, valerian, tryptophan, tiagabine.

Class / drugs	Key points
Z-drugs (non-benzodiazepine BZRAs): **zolpidem**, **eszopiclone**, **zaleplon**	**Boxed: complex sleep behaviors** (sleepwalking, sleep-driving, cooking) → stop if they occur • **next-day impairment** (no driving) • take **immediately before bed** with **7–8 hours** available • **not with/after a heavy meal** (delays onset) • **zolpidem: lower dose in women and elderly** (5 mg IR, 6.25 mg ER) • **eszopiclone:** unpleasant/**metallic taste**; 1 mg start (max 2 mg elderly/hepatic) • **zaleplon:** very short (onset only) • Schedule IV • falls (Beers)
Orexin receptor antagonists (DORAs): **suvorexant**, **lemborexant**, **daridorexant**	Onset and **maintenance** • somnolence, next-day impairment, **sleep paralysis**, hypnagogic hallucinations, cataplexy-like symptoms • **CI: narcolepsy** • **CYP3A4** (avoid strong inhibitors) • Schedule IV • suvorexant approved for insomnia in Alzheimer's
Ramelteon (melatonin MT1/MT2 agonist)	**8 mg** 30 min before bed • **sleep onset** • **no abuse**, not controlled • **CI with fluvoxamine** (CYP1A2) • avoid high-fat meal • ↑ prolactin, ↓ testosterone (rare)
Low-dose doxepin (Silenor)	**3–6 mg** • H1 antagonist at low dose → **sleep maintenance** • OK in **elderly** at this dose • avoid within 3 h of a meal • MAOIs CI
Benzodiazepines: **temazepam**, triazolam, estazolam, flurazepam, quazepam	Effective but **tolerance, dependence**, withdrawal, falls, cognitive impairment • **flurazepam** (very long) → avoid • **triazolam**: short, rebound insomnia, amnesia; CYP3A4 (azoles CI) • not first choice; avoid in elderly
OTC antihistamines: diphenhydramine, doxylamine	**Anticholinergic** (confusion, urinary retention, constipation), next-day drowsiness, **tolerance** in days • **avoid in elderly** (Beers) • short-term only
Melatonin	0.5–5 mg • weak effect on chronic insomnia • useful for **jet lag**, **shift work**, circadian disorders • supplement (quality varies)
Trazodone	Widely used off-label (25–100 mg) • orthostasis, priapism • not recommended by AASM

Special populations

- **Elderly:** CBT-I first • preferred: **low-dose doxepin**, **ramelteon**, DORAs, **low-dose Z-drugs** • **avoid** benzodiazepines, **diphenhydramine**, long-acting agents (Beers) — falls, fractures, confusion.
- **Pregnancy:** CBT-I and sleep hygiene • doxylamine or diphenhydramine occasionally • avoid benzodiazepines.
- **Substance use or sleep apnea:** avoid benzodiazepines and Z-drugs (respiratory depression) • ramelteon, doxepin.

Other sleep disorders (brief)

Disorder	Treatment
Narcolepsy	**Modafinil** 200 mg AM, **armodafinil** (↓ hormonal contraceptives — CYP3A4 induction; rash/SJS), **solriamfetol**, **pitolisant**, stimulants • **sodium oxybate** (cataplexy; boxed: CNS/respiratory depression, abuse; REMS)
Restless legs syndrome	Check **ferritin** → **iron** if < 75 ng/mL • **gabapentin enacarbil**, gabapentin, **pregabalin** (first-line) • dopamine agonists (pramipexole, ropinirole, rotigotine) → risk of **augmentation** • avoid antihistamines, antidopaminergics, SSRIs if possible
Obstructive sleep apnea	**CPAP** • weight loss (**tirzepatide** approved for OSA with obesity) • avoid sedatives
Shift work disorder	Modafinil/armodafinil, melatonin, light exposure

Insomnia drug summary

Drug	Usual adult dose	Key ADRs	Key DDIs / notes
Zolpidem IR	5 mg (women, elderly) • 5–10 mg (men)	**Complex sleep behaviors**, next-day impairment, amnesia	CYP3A4 • empty stomach
Zolpidem ER	6.25 mg (women, elderly) • 6.25–12.5 mg (men)	Same	Swallow whole
Zolpidem SL low-dose	1.75 mg (women) / 3.5 mg (men) at night awakening	Same	Only if ≥ 4 h left
Eszopiclone	1 mg → max 3 mg (2 mg elderly/hepatic)	**Metallic taste**, complex sleep behaviors	CYP3A4 inhibitors → max 2 mg
Zaleplon	5–10 mg (max 20)	Complex sleep behaviors	Sleep onset • cimetidine ↑
Suvorexant	10 mg → max 20 mg	Somnolence, sleep paralysis	**CI narcolepsy** • strong CYP3A4 inhibitors avoid
Lemborexant	5 → 10 mg	Somnolence	CYP3A4
Daridorexant	25–50 mg	Headache, somnolence	CYP3A4
Ramelteon	8 mg	Dizziness, somnolence	**CI fluvoxamine** • no abuse
Doxepin (low dose)	3–6 mg	Somnolence	MAOIs CI • cimetidine ↑
Temazepam	7.5–30 mg	Dependence, falls	LOT (no active metabolites)
Triazolam	0.125–0.25 mg	Rebound insomnia, amnesia	**CI strong CYP3A4 inhibitors**
Diphenhydramine	25–50 mg	Anticholinergic, tolerance	Avoid elderly
Melatonin	0.5–5 mg	Headache, drowsiness	Jet lag, circadian
Trazodone	25–100 mg	Orthostasis, priapism	Off-label
Modafinil	100–200 mg AM	Headache, insomnia, rash/SJS	**↓ hormonal contraceptives** (CYP3A4 inducer)

6. Substance Use Disorders (SUD)

Diagnosis (DSM-5-TR)

- **≥ 2 of 11 criteria within 12 months** → impaired control (larger amounts/longer, unable to cut down, much time spent, **craving**), social impairment (failure to meet obligations, social problems, giving up activities), risky use (hazardous situations, continued despite problems), pharmacologic (**tolerance**, **withdrawal**).
- **Severity: 2–3 mild • 4–5 moderate • ≥ 6 severe**.
- **Tolerance and withdrawal do NOT count** when the drug is taken as prescribed under medical supervision (physical dependence ≠ addiction).
- **FDA-approved drug therapy exists for: alcohol**, **opioid**, and **tobacco** use disorders only.

Alcohol withdrawal

Time after last drink	Features
6–24 hours	Tremor, anxiety, insomnia, sweating, tachycardia, hypertension, nausea
12–48 hours	Alcoholic hallucinosis (usually visual, with clear sensorium)
24–48 hours	**Withdrawal seizures** (generalized)
48–96 hours	**Delirium tremens**: confusion, agitation, fever, severe autonomic instability → can be fatal

- **Assess severity: CIWA-Ar** score (treat if ≥ 8–10; > 15 severe).
- **Benzodiazepines = first-line** (symptom-triggered dosing preferred): **chlordiazepoxide** 50–100 mg, **diazepam** 10–20 mg, or **lorazepam** 2–4 mg (**lorazepam/oxazepam** in **liver disease** and elderly) • **phenobarbital** is an alternative.
- **Thiamine 100 mg BEFORE glucose** (prevents **Wernicke encephalopathy**: confusion, ataxia, ophthalmoplegia → high-dose IV thiamine, e.g., 500 mg TID) • **folate**, **multivitamin**, **magnesium**, potassium, fluids.
- Adjuncts: dexmedetomidine (ICU), haloperidol for hallucinations (**not alone** — lowers seizure threshold) • mild outpatient: carbamazepine, gabapentin.

Alcohol use disorder: maintenance therapy

Drug	Dose	Key points
Naltrexone (opioid antagonist)	**50 mg PO daily • 380 mg IM monthly** (Vivitrol, gluteal)	**↓ craving and heavy drinking** • can start while still drinking • **CI: current opioid use** (precipitated withdrawal; need **7–10 days opioid-free**), **acute hepatitis/liver failure** • ADRs: nausea, headache, **hepatotoxicity** (LFTs), injection-site reactions • blocks opioid analgesics
Acamprosate	**666 mg TID** (2 × 333 mg)	Maintains **abstinence** (start after detox) • **renal elimination**: CrCl 30–50 → 333 mg TID; **CrCl ≤ 30 contraindicated** • **safe in liver disease** • ADRs: **diarrhea**, anxiety, depression/suicidality
Disulfiram	**250 mg daily** (125–500 mg)	Inhibits **aldehyde dehydrogenase** → acetaldehyde ↑ → flushing, nausea, vomiting, headache, hypotension if alcohol consumed • start ≥ **12 h after last drink** • avoid alcohol (including mouthwash, cough syrups) **up to 2 weeks** after stopping • **never give without patient knowledge** • hepatotoxicity • ↑ warfarin, phenytoin • **metronidazole** CI
Off-label	Topiramate, gabapentin, baclofen (liver disease), varenicline	—

Opioid intoxication and overdose

- **Triad: miosis** (pinpoint pupils), **respiratory depression**, **CNS depression**.
- **Naloxone:** 0.4–2 mg IV/IM/SC, repeat every 2–3 min • **4 mg intranasal** (available **OTC**), 8 mg nasal • **nalmefene** 2.7 mg nasal (longer acting) • naloxone has a **short half-life** → monitor for re-sedation (long-acting opioids, fentanyl) • may precipitate withdrawal.
- **Prescribe/dispense naloxone** to everyone with OUD or on high-dose opioids, and to family members.

Opioid withdrawal

- **Symptoms: mydriasis**, yawning, **lacrimation**, **rhinorrhea**, piloerection ("goosebumps"), sweating, **diarrhea**, abdominal cramps, nausea/vomiting, myalgias, anxiety, insomnia, tachycardia • uncomfortable but **rarely life-threatening** • scored with **COWS**.
- **Symptomatic treatment: clonidine** 0.1–0.2 mg q4–6h or **lofexidine** 0.18 mg × 3–4 tablets QID (FDA-approved; hypotension, bradycardia, QT) • **loperamide** (diarrhea), **ondansetron**, NSAIDs/acetaminophen, methocarbamol, hydroxyzine.
- **Best:** start **medication for OUD** (buprenorphine or methadone).

Opioid use disorder: medications (MOUD)

Drug	Key points
Methadone (full μ-agonist)	For OUD **only through certified opioid treatment programs (OTPs)** • start **20–30 mg** (max **40 mg on day 1**), increase slowly (5–10 mg every 3–5 days) → maintenance 60–120 mg • **long, variable half-life** → accumulation and **overdose risk in the first 2 weeks** • **QT prolongation** (ECG) • respiratory depression • **CYP3A4/2B6**: inducers (**rifampin**, **efavirenz**, carbamazepine, phenytoin) → **withdrawal**; inhibitors → toxicity
Buprenorphine (partial μ-agonist, κ-antagonist)	**Ceiling effect** on respiratory depression (safer) • any DEA-registered prescriber (X-waiver removed 2023) • **start only when in mild–moderate withdrawal** (COWS ≥ 8–12; ≥ 12 h after short-acting opioids, 24–72 h after long-acting/methadone) → otherwise **precipitated withdrawal** (high receptor affinity) • low-dose (micro-) induction alternative • **buprenorphine/naloxone SL** 2–4 mg → **8–16 mg day 1** → maintenance 16–24 mg/day • **naloxone** in the combination deters IV misuse (not absorbed SL) • ER injections: **Sublocade** (monthly SC), **Brixadi** (weekly/monthly) — **boxed: serious harm or death if given IV** • ADRs: constipation, headache, nausea, hepatotoxicity • CYP3A4
Naltrexone ER (Vivitrol)	**380 mg IM monthly** • must be **opioid-free 7–10 days** (detox first) • no abuse potential • **↑ overdose risk** if opioids are used after stopping (lost tolerance) • hepatotoxicity, injection-site reactions

- **Pregnancy: methadone** or **buprenorphine** (mono or combination product) — **do not** detox during pregnancy • infant: neonatal opioid withdrawal syndrome (manageable).
- Continue MOUD long term • combine with counseling • MOUD ↓ overdose deaths.

Tobacco use disorder

- **Varenicline** (most effective single agent), **combination NRT** (patch + short-acting), **bupropion SR** • combinations (varenicline + NRT) for heavy smokers • see the Asthma/COPD file for full dosing.
- **Quitting smoking ↑ levels of CYP1A2 substrates** (clozapine, olanzapine, theophylline, caffeine) → monitor.

Other substances

Substance	Key points
Stimulants (cocaine, methamphetamine)	**No FDA-approved medication** • **contingency management** (most effective) • intoxication: agitation, hypertension, tachycardia, hyperthermia, chest pain → **benzodiazepines** • cocaine chest pain: benzodiazepines, aspirin, nitroglycerin, CCBs • **avoid beta-blockers acutely** (unopposed alpha stimulation) • off-label: bupropion + naltrexone for methamphetamine
Cannabis	No approved medication • **cannabinoid hyperemesis syndrome** (cyclic vomiting relieved by hot showers) → stop cannabis, **haloperidol**, **topical capsaicin**
Benzodiazepines	Withdrawal can cause **seizures** → **slow taper** (switch to a long-acting agent like diazepam or clonazepam)
Gabapentinoids	Misuse (especially with opioids) → respiratory depression

Substance use drug summary

Drug	Usual adult dose	Key ADRs	Key DDIs / notes
Chlordiazepoxide	50–100 mg PO, symptom-triggered (CIWA)	Sedation, respiratory depression	Long-acting • avoid in liver disease
Lorazepam	1–4 mg PO/IV, symptom-triggered	Sedation	**Preferred in liver disease**, elderly
Diazepam	10–20 mg, symptom-triggered	Sedation	Active metabolites
Phenobarbital	Loading/fixed protocols	Sedation, respiratory depression	Alternative to benzodiazepines
Thiamine	100 mg daily (500 mg IV TID if Wernicke's)	Rare	**Give before glucose**
Naltrexone	50 mg PO daily • 380 mg IM monthly	Nausea, **hepatotoxicity**	**CI with opioids** (7–10 days opioid-free) • blocks opioid analgesia
Acamprosate	666 mg TID	**Diarrhea**	**Renal**: CrCl 30–50 → 333 mg TID; ≤ 30 CI • OK in liver disease
Disulfiram	250 mg daily	Hepatotoxicity, metallic taste, neuropathy	**Disulfiram-alcohol reaction** • ↑ warfarin, phenytoin • CI metronidazole
Naloxone	0.4–2 mg IV/IM q2–3 min • 4 mg nasal	Precipitated withdrawal	Short half-life → re-sedation
Nalmefene	2.7 mg nasal	Precipitated withdrawal	Longer acting
Methadone	20–30 mg start (max 40 mg day 1) → 60–120 mg	**QT**, respiratory depression, constipation	**CYP3A4/2B6**: rifampin, efavirenz ↓ (withdrawal) • other QT drugs
Buprenorphine/naloxone	SL 2–4 mg → 8–16 mg day 1 → 16–24 mg/day	Constipation, headache, **precipitated withdrawal** if started too early	CYP3A4 • ceiling on respiratory depression
Buprenorphine ER (Sublocade, Brixadi)	Monthly SC (Sublocade) • weekly/monthly (Brixadi)	Injection-site reactions	**Boxed: never IV**
Naltrexone ER (Vivitrol)	380 mg IM monthly	Injection-site reactions, hepatotoxicity	Opioid-free 7–10 days • overdose risk after stopping
Clonidine	0.1–0.2 mg q4–6h	**Hypotension**, sedation, rebound HTN	Opioid withdrawal (off-label)
Lofexidine	0.54 mg (3 × 0.18 mg) QID × up to 14 days	Hypotension, bradycardia, **QT**	Taper over 2–4 days
Varenicline	0.5 mg daily × 3 d → 0.5 mg BID × 4 d → 1 mg BID	Nausea, vivid dreams	Renal adjustment
Bupropion SR	150 mg daily × 3 d → 150 mg BID	Seizures, insomnia	CI seizures, eating disorders, MAOIs
Nicotine replacement	Patch 21/14 mg • gum/lozenge 2–4 mg	Skin irritation, vivid dreams	Combination NRT most effective

Exam Traps and Frequently Tested Points

Schizophrenia

1. **≥ 2 symptoms for 1 month** (one = delusions, hallucinations, or disorganized speech) + **6 months** total.
2. **High-potency FGAs** → EPS • **low-potency** → sedation, anticholinergic, orthostasis.
3. **Risperidone/paliperidone** → prolactin • **clozapine/olanzapine** → metabolic • **aripiprazole** → akathisia, ↓ prolactin.
4. **Ziprasidone** (≥ 500 kcal) and **lurasidone** (≥ 350 kcal) → take with food • ziprasidone → QT.

5. **Clozapine** → treatment-resistant (≥ 2 failed trials) • neutropenia (ANC), seizures, myocarditis, constipation • **quitting smoking ↑ levels**.
6. **Dystonia → diphenhydramine/benztropine IM • akathisia → propranolol • TD → valbenazine/deutetrabenazine** (anticholinergics worsen TD).
7. **NMS:** fever, lead-pipe rigidity, ↑ CK → stop drug, dantrolene/bromocriptine.
8. **Boxed:** ↑ mortality in elderly with dementia.
9. **Metabolic monitoring** at baseline, 12 weeks, then yearly.

MDD

1. **SIG E CAPS**, ≥ 5 symptoms for ≥ 2 weeks • **screen for bipolar** before antidepressants.
2. **Trial 4–8 weeks** • continue **4–9 months after remission** • ≥ 3 episodes → long term.
3. **Boxed suicidality < 25 years**.
4. **Citalopram max 40 mg (20 mg if > 60)** — QT.
5. **Paroxetine** → anticholinergic, worst discontinuation, avoid pregnancy, ↓ tamoxifen.
6. **Fluoxetine** → long half-life, 5-week MAOI washout.
7. **Fluvoxamine + tizanidine** → contraindicated (CYP1A2).
8. **Bupropion** → no sexual dysfunction/weight gain • CI seizures, eating disorders.
9. **Mirtazapine** → sedation, weight gain • **trazodone** → priapism.
10. **Venlafaxine** → ↑ BP • **duloxetine** → hepatotoxicity.
11. **TCAs** → lethal overdose (sodium bicarbonate), avoid elderly.
12. **MAOIs** → tyramine, sympathomimetics, serotonergic drugs (linezolid, meperidine, dextromethorphan) • 2-week washout.
13. **Serotonin syndrome** (rapid, clonus, hyperreflexia) vs **NMS** (slow, rigidity) • cyproheptadine.
14. **Esketamine** → REMS, 2-hour observation • **zuranolone** → postpartum depression, no driving 12 h.
15. **Pregnancy → sertraline • children → fluoxetine (≥ 8), escitalopram (≥ 12)**.

Bipolar disorder

1. **Mania ≥ 1 week**, **hypomania ≥ 4 days** • DIGFAST • bipolar I = mania; bipolar II = hypomania + depression.
2. **Avoid antidepressant monotherapy**.
3. **Lithium levels:** acute 0.8–1.2, maintenance 0.6–1.0 • trough 12 h, 5 days after change.
4. **Lithium ↑ by thiazides, ACEi/ARBs, NSAIDs, dehydration** • ↓ by caffeine, theophylline.
5. **Lithium ADRs:** tremor, nephrogenic DI, hypothyroidism, Ebstein's anomaly, CKD.
6. **Valproate** → mixed/rapid cycling • avoid in women of childbearing potential.
7. **Lamotrigine** → maintenance (prevents depression), not acute mania.
8. **Bipolar depression:** quetiapine, lurasidone, cariprazine, lumateperone, olanzapine/fluoxetine.

Anxiety

1. **SSRIs/SNRIs + CBT** first-line • start low in panic • 8–12 weeks for full effect.
2. **Buspirone** → GAD only, delayed onset, not PRN, no dependence.
3. **Benzodiazepines** → short-term bridge • boxed: opioids, abuse, dependence • taper slowly.
4. **LOT** (lorazepam, oxazepam, temazepam) → liver disease and elderly.
5. **Performance anxiety → propranolol • PTSD nightmares → prazosin • avoid benzodiazepines in PTSD**.
6. **OCD** → higher SSRI doses, 8–12 weeks, clomipramine.

Insomnia

1. **CBT-I** first-line.
2. **Z-drugs** → complex sleep behaviors (boxed) • **zolpidem 5 mg in women/elderly** • eszopiclone metallic taste.
3. **DORAs** (suvorexant) → maintenance; CI narcolepsy.
4. **Ramelteon** → onset, no abuse • CI fluvoxamine.
5. **Low-dose doxepin** → maintenance, OK in elderly.
6. **Avoid diphenhydramine and benzodiazepines in elderly**.

7. **Trazodone, melatonin, OTC antihistamines** → not recommended by AASM for chronic insomnia.

Substance use disorders

1. **≥ 2 of 11 criteria** • 2–3 mild, 4–5 moderate, ≥ 6 severe • tolerance/withdrawal don't count if prescribed.
2. **Alcohol withdrawal → benzodiazepines** (CIWA) • **thiamine before glucose** • lorazepam/oxazepam in liver disease.
3. **Delirium tremens** 48–96 h • seizures 24–48 h.
4. **Naltrexone** → CI with opioids; hepatotoxicity • **acamprosate** → renal, OK in liver disease • **disulfiram** → no alcohol 2 weeks after.
5. **Opioid overdose → naloxone** (short half-life, re-sedation).
6. **Buprenorphine** → start in mild–moderate withdrawal (else precipitated withdrawal) • ceiling effect.
7. **Methadone** → OTP only for OUD, QT, overdose risk early, CYP3A4/2B6 inducers ↓ levels.
8. **Naltrexone ER** → 7–10 days opioid-free.
9. **Lofexidine/clonidine** → opioid withdrawal symptoms.
10. **Pregnancy OUD → methadone or buprenorphine**.
11. **Stimulants and cannabis** → no approved drugs • cocaine: avoid acute beta-blockers.

> **Note:** Doses and guideline recommendations differ between references and are updated often. **Use the numbers in the reference you are studying for the exam.**

Good luck on your exam!